

YOHANNES BELAI

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PROFESSIONAL SUMMARY

Result-driven Software Engineer with expertise in backend architecture (Java, Python, Node.js) and cloud infrastructure (AWS). Proven track record of optimizing data pipelines reducing execution time by 35% and query latency by 60%. Experienced in deploying full-stack AI-integrated applications and mentoring aspiring developers.

EXPERIENCE

Primary Teacher | *Mobile App Building with AI* Jul 2026 - Aug 2026

- Directed a comprehensive 6-week curriculum on Mobile App Building with AI for 10+ students across two distinct age cohorts (5-10 and 11-17).
- Facilitated the development of 10+ functional mobile applications using MIT App Inventor, teaching foundational logic and AI integration concepts.

Teaching Assistant | *Summer Coding Camp (Intermediate Python Track)* Jun 2026 - Early Aug 2026 (Mondays)

- Mentored 10+ high school students through an intensive 5-week Python track, covering OOP and algorithm design.
- Designed and delivered weekly lesson plans, improving student project completion rates by 30%.
- Conducted code reviews for 10+ student projects, providing targeted feedback on design patterns and code clarity.
- Debugged live student issues during lessons, effectively resolving 10+ coding blockers weekly under time pressure.

Software Developer (Contract) | *Alignerr* Aug 2025 - Present | Remote

- Built asynchronous data pipeline using AWS Lambda, SQS, and S3 processing 50M+ daily records for large-scale AI model training; reduced job execution time by 35%.
- Refactored legacy Java codebase across 15+ modules for AI training workflows, decreasing production runtime errors by 40% and improving maintainability.
- Optimized database queries and added strategic indexing across MongoDB collections, reducing query latency by 60% and accelerating downstream workflows.
- Validated and refactored 10+ lines of AI-generated code weekly for production readiness, ensuring adherence to clean architecture principles.
- Collaborated within an Agile environment with a distributed team of 8 engineers using Git workflows.

PROJECTS

Drill | *AI-Powered Study Material Generator (React + Spring Boot + Google Gemini API)* Live: drillapi.onrender.com

- Engineered a full-stack study platform capable of processing multi-file PowerPoint uploads to generate JSON-formatted flashcards and quizzes via the Google Gemini API.
- Built Spring Boot REST API with Apache POI for robust .pptx parsing, extracting structured data from files up to 50MB in size.
- Implemented resilient custom retry logic for LLM requests, achieving an API success/uptime rate of 99%.
- Deployed backend on Render and frontend (React/Vite SPA) on Vercel, managing environment variables across staging and production.

SyllabiXtract | *AI-Powered Syllabus Processing Platform (React + Auth0 + Node.js Backend)*

- Implemented React frontend (Vite + Tailwind CSS) supporting multi-file async upload UX with progress indicators, built for 10+ simulated users.
- Integrated OAuth2 authentication via Auth0 and collaborated with backend engineers on REST API contracts and request/response schemas.

Campus Events API | *RESTful Backend for Event Management* **Deployed:** campusevents-olmp.onrender.com/api
(Node.js + Express + MongoDB)

- Designed and shipped 12 RESTful endpoints with role-based access control (admin, organizer, user) for 10+ concurrent user profiles.
- Implemented JWT-based authentication with bcrypt password hashing and a secure token refresh mechanism.
- Modeled MongoDB schema with optimized indexing for complex queries (events by date, user registrations), reducing search latency.

SKILLS

Languages: Java, Python, JavaScript, SQL

Backend & Frameworks: Spring Boot (REST APIs, JPA/Hibernate ORM), Spring Security, Node.js, Express.js, JWT, OAuth2, Apache POI

Frontend: React, Vite, Tailwind CSS, HTML/CSS

Databases: PostgreSQL, MongoDB (indexing, schema design)

Cloud & DevOps: AWS (Lambda, SQS, S3), Docker, Git/GitHub, CI/CD basics, Render, Vercel

AI & Tools: Google Gemini API, LlamaParser, prompt engineering, AI-assisted code review

EDUCATION

Towson University | *B.S. in Computer Science*

Montgomery College | *A.S. in Computer Science*